

REVIEW

Diabetes and advanced glycation endproducts

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Bio-reactive advanced glycation endproducts (AGE) alter the structure and function of molecules in biological systems and increase oxidative stress. These adverse effects of both exogenous and endo-

genously derived AGE have been implicated in the pathogenesis of diabetic complications and changes associated with ageing including atherosclerosis, renal, eye and neurological disease. Specific AGE receptors and nonreceptor mechanisms contribute to these processes but also assist in the removal and degradation of AGE. The final disposal of AGE depends on renal clearance. Promising pharmacologic strategies to prevent AGE formation, reduce AGE toxicity, and/or inactivate AGE are under investigation.

Introduction

Prolonged exposure to hyperglycaemia is now recognized as the primary causal factor in the majority of diabetic complications [1, 2]. Indeed, glucose has a wide range of transient and reversible effects on cell function [3, 4] as well as effects that are irreversible and can cause progressive, cumulative dysfunction [5]. This suggests that persistent, rather than acute, metabolic alterations are of pivotal importance in the development and progression of diabetic complications. Amongst the irreversible changes which occur as a direct result of hyperglycaemia is the formation of advanced glycation endproducts (AGEs) via the Maillard reaction. AGEs have a range of chemical, cellular, and tissue effects and act as mediators, not only of diabetic complications, but also of widespread changes associated with ageing.

The following review focuses on the role that advanced glycation plays in the initiation and progression of diabetic complications. We will also review agents which can ameliorate the toxic effects of these products and speculate about novel therapeutic interventions.

AGE: metabolic aspects

Endogenous AGE

Effects of short-lived AGE molecules. Advanced glycation endproduct formation has been known to have significant effects on macromolecular structure and function. Until recently it was thought that AGE formation involves primarily long-lived extracellular proteins and occurs as a function of time, thus representing a form of molecular senescence. It is now clear that AGEs arise on short-lived molecules

as well, including circulating plasma proteins and lipids, and that their levels are significantly elevated in diabetic patients and in patients with impaired renal clearance [6, 7]. It is also recognized that they can form rapidly on cytoplasmic proteins and nucleic acids. Indeed, it has been demonstrated that intracellular AGEs may form at a rate up to 14-fold faster in high (30 mM) glucose conditions [8]. Such AGE-modification of short-lived molecules is known to involve oxidation of proteins and lipids, to disrupt molecular conformation, to alter enzymatic activity, to reduce degradative capacity, and to result in abnormal recognition and clearance by receptors [9–12].

The broad pathological significance of AGE-modification is best reflected in the glycation of lipids and lipoproteins, as in the case of apoprotein B (ApoB) and low density lipoprotein (LDL) [13]. Dyslipidaemic changes are evident in diabetic patients and are characterized by increased levels of LDL which greatly predispose these patients to atherosclerosis, with subsequent increased risk for coronary heart disease and stroke. Advanced glycation of the lipid component of LDL occurs concomitantly with LDL oxidation *in vitro* [13]. The presence of amino groups on certain phospholipids such as phosphatidylethanolamine and phosphatidylserine provides appropriate sites with which glucose can react with lipid amines to form AGEs [14]. During glycation, fatty acid residues can be oxidized independently of transition metals or exogenous free radical generating systems [14]. Significantly, LDL oxidation follows the formation of AGE-LDL, whilst both occur in direct proportion to glucose concentration and are inhibited in the presence of the AGE inhibitor aminoguanidine [14]. Thus, it is apparent that free amine:glucose interactions with lipids are spontaneous and natural *in vivo* steps leading to fatty acid glycation and oxidation products.

The ApoB component of LDL is a relatively large protein with many potential lysine and arginine AGE modification sites, although the predominant site of such modification has been found distally to the N-terminus of the LDL-receptor binding domain [13]. AGE-ApoB levels are approximately 4-fold higher in diabetic patients [14, 15]. The pathophysiological implications of this AGE-modification have been demonstrated in a study in which AGE-LDL was injected into transgenic mice expressing the human LDL receptor. The clearance of the modified

LDL was delayed compared with that of native LDL [15]. This suggests that advanced glycation of ApoB can lead to hyperlipoproteinaemia and, thus, may actively contribute to atherosclerosis by reducing LDL clearance and by facilitating AGE-LDL deposition in the vessel wall via AGE-receptor interactions (see below) [16, 17].

Effects of long-lived AGE molecules. The rate of formation of AGEs exceeds that predicted by first-order kinetics. This implies that, over time, even modest hyperglycaemia can result in significant accumulation of AGEs on long-lived macromolecules [18–20]. This is well illustrated on certain long-lived proteins such as those of the ocular lens. For example, the progressive post-translational modification of lens crystallins by glucose-derived AGEs explains the premature lenticular browning and cumulative crosslinking occurring during the course of diabetes [21, 22]. This accounts for a significant proportion of lenticular opacification and subsequent cataract formation during ageing as well as in diabetes [23, 24]. These changes can be reproduced *ex vivo* in the presence of high glucose concentrations and can be effectively prevented in the presence of the AGE-inhibitor aminoguanidine [25]. Furthermore, the collagen network of human vitreous gel contains increased levels of AGEs in diabetics [25] and vitreous AGE levels show a significant correlation with age, suggesting that AGEs play a major role in diabetes- and age-related vitreous alterations including progression of retinopathy and posterior vitreal detachments (PVDs) [25].

Proteins constituting the extracellular matrix (ECM) and vascular basement membranes (BM), which are amongst the longest lived in the body, are highly susceptible to AGE-modification. Functionally, AGE-mediated crosslinks in BM are known to cause reduced solubility and decreased enzymatic digestion [26, 27]. In addition, AGE formation has been shown to impair the geometrically ordered self-assembly of BM proteins, thereby causing structural and functional abnormalities. For example, AGE-modification of laminin, vitronectin, and collagen can seriously alter molecular charge characteristics, upset the ability to form precisely assembled three dimensional matrix aggregates, and thus disrupt biological attachment sites which enable cells to adhere to their substrates

[11, 28–30]. Moreover, the reduced binding of heparin sulphate proteoglycan matrix to AGE-modified collagen, laminin, and fibronectin significantly alters the polyanionic nature of BM [31, 32], seriously affecting the charge-mediated properties of BM. Thus, the very presence of AGE on vascular BM may have dire pathological consequences, particularly in diabetics, who have an accelerated accumulation of AGEs.

AGE-DNA and embryopathy. Whilst the primary amino groups of nucleotides are less reactive nucleophiles than are ϵ -amino groups of lysine and arginine, nucleic acids can, nonetheless, react with reducing intracellular sugars to form Amadori and AGE-products with characteristic fluorescence and spectral properties [33]. AGE-formation on DNA can cause single strand breaks in genomic DNA which can have serious teratogenic effects [33–35].

Teratogenic effects may also be promoted in diabetes by enhanced glycation of histone proteins, which have a vital structural role in maintaining nucleosomes and hence DNA integrity. Intracellular sugars such as glucose-6-phosphate and ADP-ribose react strongly with amino groups on histones where they can cause crosslinking *in vitro* [36, 37] as well as *in vivo* [38].

Maternal diabetes has been associated with congenital malformations and increased foetal mortality and morbidity [39]. It is estimated that there is a two to three fold increased incidence of perinatal infant fatalities as a result of congenital malformations in the offspring of women with insulin dependent diabetes compared with those born to nondiabetic mothers [40]. Most studies show a generalized increase in malformations involving multiple organ systems, although the teratogenic mechanisms are largely unknown. From the available evidence it would appear that AGE-mediated DNA damage could be a significant factor in the teratogenicity occurring in diabetic pregnancies. The incidence of congenital abnormalities is not increased in short-term, pregnancy-induced hyperglycaemia, as it is when long-term poor glycaemic control predates conception [41]. This observation, together with the marked reduction in foetal abnormalities with effective management of hyperglycaemia during pregnancy has added further evidence implicating the role of AGEs in diabetic embryopathy.

Major exogenous sources of AGEs

Diet. Recent studies suggest that AGEs introduced to biological systems from exogenous sources such as diet and smoking may have significant impact on disease mechanisms. Diet is the major source of exogenous AGE, with the highest content in complex foods, such as those rich in carbohydrates and fats. Formation of AGE is enhanced by exposure to heat; thus AGE content, which is responsible for the browning of food as it cooks, increases with cooking temperature and duration. Whilst the abundance of dietary AGEs including methylglyoxal (MG) and carboxymethyl lysine (CML) has long been recognized, their significance as potential toxins was not appreciated as their absorption was estimated to be only approximately 10% of that ingested [42].

The availability of new AGE-specific bioassays has facilitated the study of the dietary content, bioavailability and renal elimination of these moieties, schematically depicted in Fig. 1. Koschinsky et al. [43], showed that AGE immunoreactivity can increase by 200-fold in egg-whites cooked with fructose, compared with egg-whites prepared in the identical manner in the absence of fructose. This study also confirmed the absorption of 10% of ingested AGE and showed that only one-third of that absorbed is excreted within 48 h in the urine of patients with normal renal function [43]. As AGEs that are not cleared by the kidney are distributed to the tissues where they remain biologically active [43], the data imply that dietary AGEs may pose a significant environmental risk, particularly to patients with nephropathy.

The contribution of dietary AGEs to injury *in vivo* is apparent from a study in which mice were randomized to either a high or low AGE diet, both isocaloric and identical in protein and nutrient content; mice on the high AGE diet exhibited significant albuminuria as well as microscopic evidence of glomerular hypertrophy and/or sclerosis, effects not seen in the mice fed the low AGE diet (Fig. 2) [44].

Other *in vivo* evidence of the role of dietary glycotoxins was obtained from a study which maintained ApoE-deficient mice on chow which was either high or low in AGE content for a total of 5 weeks. After 1 week, a femoral artery denudation injury was induced and the designated diet was maintained for another 4 weeks before the injured arteries were

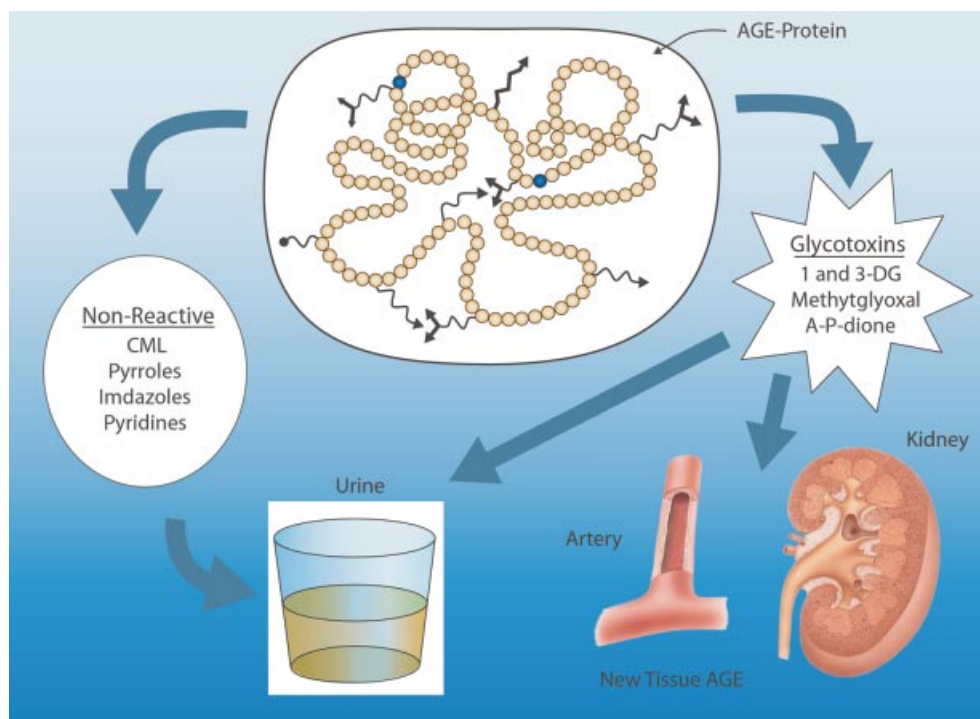


Fig. 1 Schematic representation of the fate of diet-derived AGEs. Cooked foods contain sugar-derived protein or lipid glycation intermediates that may include noncrosslinking products, such as *N*-ε-carboxymethyl-lysine (CML), pyrroles, imidazoles, pyridines (left insert), or crosslink forming, reactive intermediates (glycotoxins), such as 1-,3-deoxyglycozone (3-DG), methylglyoxal, protein-linked A-P-dione (right insert). The former are presumably readily excreted in urine, whilst the latter may reattach onto serum or tissue components to form new AGEs with the eventual pathological consequences.

examined. Injured arteries in the animals fed with the low-AGE diet showed a significant reduction in neointimal area (Fig. 3). In addition, those animals developed less complex lesions with fewer foam cells within the neointima, associated with a 40% decrease in serum AGE levels [45].

Similarly, Apo-E deficient streptozotocin diabetic mice which were fed with a diet with AGE content 10-fold lower than that of regular rodent chow had significantly suppressed aortic atherogenesis compared with animals on the regular chow. This was associated with significantly reduced serum AGE levels despite elevated lipid levels [46].

A recent study in humans compared the effects of two diets; both were compatible with American Heart Association and American Diabetes Association guidelines but they differed 6-fold in their AGE content. Eleven diabetic subjects with normal renal function were enrolled in the crossover study; they consumed each diet for a 2-week period separated by a 2-week washout. After 2 weeks on the low AGE

diet, there was a significant decrease in fasting serum AGE levels. Furthermore, blood mononuclear cell expression of $\text{TNF-}\alpha$ mRNA and serum levels of vascular cell adhesion molecule-1 (VCAM-1), which were significantly higher at the end of the high AGE diet period, decreased by approximately 30–50% at the end of the low AGE diet period [47]. The fact that reduction in dietary AGE content resulted in lower levels of inflammatory mediators in humans, as well as the animal data above [45, 46], underscore the potential role of a low AGE diet in the primary prevention of atherosclerosis as well as in the prevention of restenosis after coronary angioplasty.

Smoking. As tobacco leaves are dried in the presence of sugars, a process called curing, the Maillard reaction cascade leads to the formation of glycated and oxidized derivatives. Upon combustion, reactive AGE species are volatilized. These species, also termed 'glycotoxins', are inhaled, absorbed through the lungs and become conjugated with serum proteins,

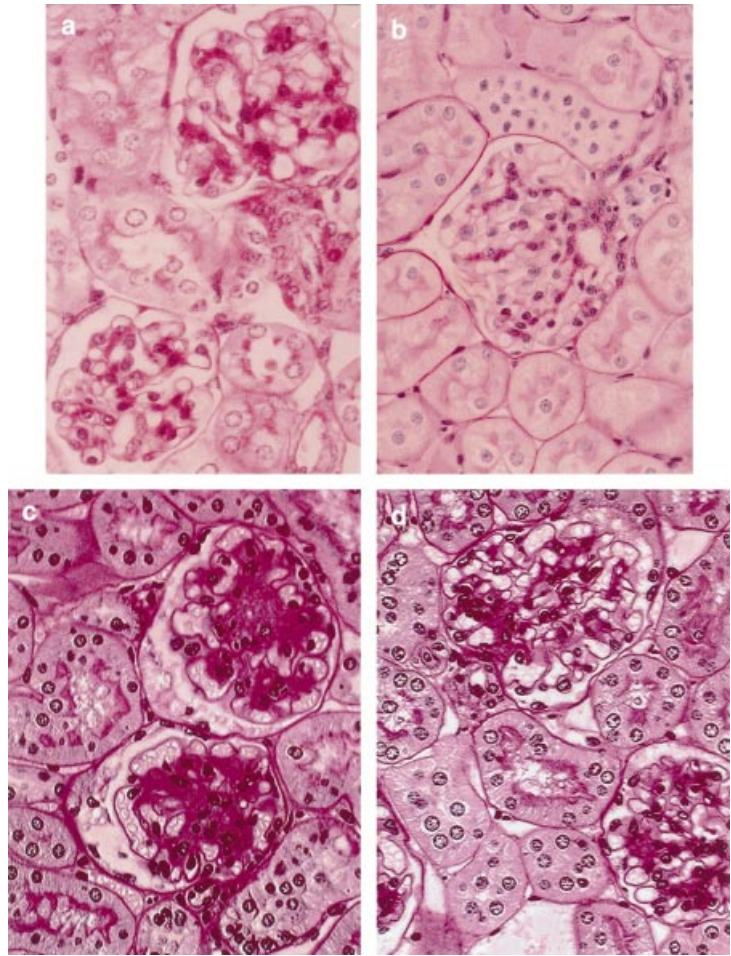


Fig. 2 Glomerular disease development is delayed in kidneys from diabetic NOD and db/db mice fed a regular diet of low AGE content. Panels a, b: db-NOD mice after 2-months on diet (H&E); Panels c, d: db/db mice after 5 months (H&E); Panels a, c: Standard mouse chow (high in AGE); Panels b, d: Same diet containing 5-fold lower AGE.

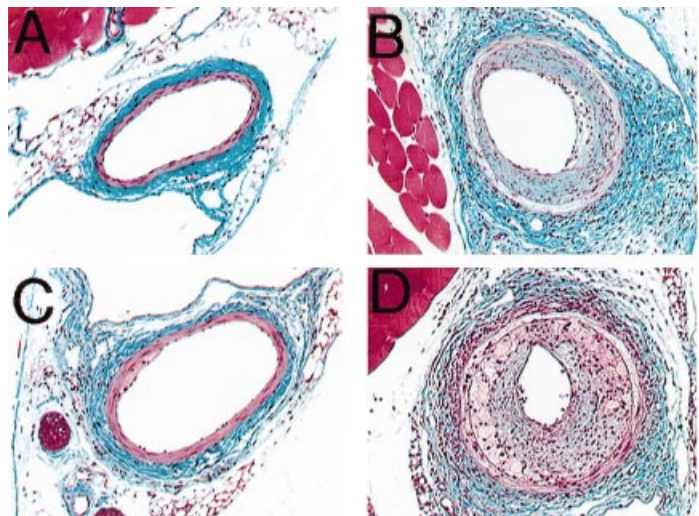


Fig. 3 Postinjury arterial restenosis is delayed in ApoE-deficient mice fed low AGE diet. Cross sections of normal mouse femoral arteries 4 weeks after injury. A and C: sham operated, uninjured controls; B and D: injured arteries. ApoE mice were fed with either low AGE diet or high AGE diet for a total of 5 weeks. Injured arteries underwent transluminal endothelial denudation within 1 week after diet (CME staining; magnification $\times 200$).

including lipoproteins [48]. This is reflected in the fact that total serum AGE and AGE-apoprotein B levels in cigarette smokers were found to be signifi-

cantly higher than in nonsmokers [48]. Also, smokers, and especially diabetic smokers, had high AGE levels in their arteries [49, 50] and ocular lens [50].

AGE: cellular interactions

Specific and nonspecific AGE-receptor systems

Early in AGE research it was speculated that a natural receptor-based system existed *in vivo* whereby AGE products could be removed from tissues, thereby limiting their deleterious effects. Work conducted over the last 15 years has led to the recognition of a complex AGE receptor system which appears to play a key role in AGE-related biology as well as in the pathology associated with the complications of diabetes and ageing [51]. It was originally demonstrated that AGE-modified proteins are recognized by specific receptors which are unrelated to previously described scavenger receptor systems [52]. In recent years, several AGE-binding molecules have been described.

Two AGE-binding proteins, a ~60 kDa (p60) and ~90 kDa (p90), isolated from mouse macrophages and rat liver membranes, were initially identified [53, 54]. P60, now referred to as AGE-R1, with characteristic membrane-spanning and signal domains, was shown to be homologous to a ~50 kDa component of the oligosaccharyltransferase complex (OST 48) [55], whilst the p90, also termed AGE-R2, with complete homology to an 80-kDa protein, was found to be a phospho-tyrosine containing protein [56] linked to early signalling via several adaptor molecules, e.g. Shc and Grb2. An additional 32 kDa AGE-binding protein with links to the AGE receptor complex (AGE-RC) was described shortly afterwards and is known as AGE-R3. This protein, also called Galectin-3, Mac-2, or carbohydrate binding protein-35, exhibits high-affinity binding for AGE ligands [57].

Evidence indicates that AGE-R1 and -R3 are largely responsible for AGE-recognition and high-affinity binding. AGE-R2 was shown to be subject to AGE-induced phosphorylation [58]. The R2 component of the AGE-R may thus play a role in signal transduction and cell activation associated with AGE receptor binding [59, 60].

Amongst the other cell-associated receptors interacting with AGEs are RAGE [61, 62], ScR-II [63, 64] and CD36 [65]. RAGE, a multiligand member of the immunoglobulin superfamily, though not as efficient in AGE endocytosis and turnover [66], is viewed increasingly as an intracellular signal-transducing,

or pro-inflammatory, peptide. In this regard, RAGE may be more accurately classified in the family of oxidant-stress inducing signalling molecules or cofactors. Consistent with the above, in animal models, brief infusion of soluble truncated RAGE is reported to intercept diverse processes such as endothelial leakage, atherosclerosis, and inflammatory bowel disease [61, 66]. In addition, specific and saturable binding of AGE-BSA to CD36-CHO cells supports the role of CD-36 as a receptor for AGE-proteins [65]. As with RAGE, this scavenger receptor system which is highly expressed on macrophages, although not restricted to AGE uptake, may contribute to AGE-mediated cellular changes, particularly in the context of atherosclerosis and foam cell formation [63, 67, 68].

A more recently discovered molecule with significant AGE-binding affinity and intriguing anti-AGE properties is the known host-defense protein Lysozyme (LZ). Lysozyme is a well-characterized, naturally occurring antimicrobial protein that exerts its effect through the catalytic degradation of the peptidoglycan component of the bacterial cell wall [69]. The AGE-binding site was mapped to a 17-amino acid-long hydrophilic domain, bound on both sides by cysteines, located within one of the two LZ catalytic regions. This AGE-binding cysteine-bounded domain is termed ABCD loop [70]. The previously unrecognized high affinity of lysozyme for AGEs ($K_d = 50$ nM) included tissue-reactive derivatives found in circulation, suggesting that LZ could be used for the capture and disposal of toxic AGEs formed *in vivo* [70, 71].

Recent reports have shown that LZ administration to nonobese diabetic (NOD) and db/db (+/+) mice normalized serum levels of AGE (sAGE), increased urinary AGE clearance, and improved albuminuria in both diabetic animal models [72, 73]. In addition, in cultured mesangial cells, LZ could suppress the AGE-enhanced expression of several important modulators of kidney structure and function such as PDGF-B, $\alpha 1$ type IV collagen, and tenascin mRNA. LZ, which appears to be expressed by mesangial cells, also normalized the AGE-suppressed MMP-9 gene expression and activity [74]. Thus, LZ revealed several novel anti-AGE properties including enhanced AGE-turnover by macrophages, suppression of AGE pro-inflammatory events, and improved renal AGE clearance. The mechanisms involved in these findings are under investigation.

AGE-receptor regulation

Different activities of the AGE-receptor system have been reported to be modulated by diabetic factors, e.g. glucose, insulin, AGEs, and reactive oxygen species (ROS) in many types of cells [75–78]. Most commonly used measures for assessing AGE-receptor modulation have centred on parameters of cell activation.

Advanced glycation endproduct-stimulated inflammatory cytokine production by macrophages and its autoregulatory impact on receptor expression and endocytic activity were amongst the original observations which broadened the concept of a system known mostly for its extracellular physico-chemical properties. Since the first findings, much more intricate bioactive mechanisms have emerged which have linked together sensors of intracellular oxidant stress and inflammatory responses to structural/mechanical tissue functions, all of which have implications in early development and growth, as well as in diverse chronic conditions throughout the body.

The link between AGE-receptor up-regulation and cellular activation has been confirmed for a number of the presently identified receptor components. However, cell activation can occur by nonreceptor pathways, or by intracellularly generated glycoxidant derivatives leading to ROS generation and oxidant stress [53, 79]. Cell activation is also induced in systemic diseases such as hyperlipidaemia, uraemia, amyloidosis, and Alzheimer's disease, all of which are associated with elevated AGE, amongst other metabolites. This makes the receptor's role much more difficult to define.

Much less is known about the regulation of receptor-dependent AGE endocytosis and catabolism. This may be attributable largely to the pleomorphic and often unstable nature of AGE-ligands. A limited number of studies have suggested either enhancement or suppression of receptor-dependent AGE-degradation in cells from diabetic or ageing animals and humans [67, 76–78].

The macrophage AGE-receptor system, which is the one most closely linked to AGE-turnover, was initially thought to include autoregulatory switches, allowing it to respond to rising AGE levels and to reduce tissue damage [76, 77]. However, the means by which this balance would be maintained *in vivo* during times of excessive AGE accumulation remains

largely unknown. As yet undefined metabolic or genetic factors could tip this balance towards pro-inflammatory events via receptor components tied to signalling, e.g. RAGE, AGE-R2, or AGE-R3, or else by receptor-independent mechanisms [53–55, 61, 67]. Alternatively, it could be that delayed AGE processing and disposal, as a result of the lack or malfunction of the endocytotic portion of the AGE-receptor, could promote cytotoxic inflammatory events, thus tipping the balance in the direction of organ damage as shown experimentally in AGE-R3 deficient mice [80, 81].

The first indication supporting this hypothesis in nature was based on findings of suppressed AGE-receptor uptake and degradation, combined with high circulating and kidney tissue AGE levels in NOD mice [82, 83]. These findings were the first to raise the importance of AGE-R1 as a site specific for cell-mediated AGE-processing and degradation. These data were also the first to suggest an *in vivo* functional impairment of receptor-dependent AGE turnover.

In support of these animal data, reduced expression of AGE-R1 in human PBM and in immortalized lymphoblasts from type 1 diabetic (T1D) patients with severe diabetic nephropathy was associated with elevated serum AGE levels and severe diabetic complications [84, 85]. These data indicate that genetic modulation of this receptor system, via means that are not yet recognized, may contribute to organ damage in complication-prone patients.

Genetic analysis of AGE-receptors

For a number of the AGE-receptor-related molecules, the genomic organization and chromosomal location [61, 86–88], as well as several prevalent gene polymorphisms [89, 90], have come to light. Recently, screening for mutations was performed in 48 T1D patients with or without nephropathy, using single strand conformational polymorphism analysis (SSCP) and direct sequencing of allelic PCR fragments [89]. Thus far, none of the polymorphisms found exhibited a clear connection to diabetic complications. However, further investigations are in progress to establish the significance of gene polymorphisms in unexplored regions of AGE-R1 and other molecules, in order to determine whether a primary or secondary genetic link exists.

Degradative mechanisms for removal of AGE-modified molecules

Although endogenous AGE formation can be moderated by lowering blood glucose levels and with the help of various natural antioxidant systems, AGEs eventually accumulate on long-lived macromolecules. Removal of existing AGE-crosslinks from tissue components is conducted largely through extracellular proteolysis and by scavenger cells such as tissue macrophages which ingest AGEs via AGE-specific or nonspecific, e.g. scavenger, receptors. It is also becoming clear that mesenchymal cells such as vascular endothelium or mesangium may also play an important role in AGE-removal. The need for efficient AGE elimination may account, in part, for the increased endocytic activity of vascular endothelium under conditions of high glucose *in vivo* and *in vitro* [91, 92]. Generally, AGE-modified molecules are recognized and internalized by cell-surface receptor-mediated endocytosis, degraded intracellularly, and subsequently released as low molecular weight AGEs, known as 'second generation AGEs' [93]. Whilst these second generation AGEs include reactive intermediates with high crosslinking or oxidative reactivity, their effects may be limited by renal excretion [6, 7, 94–96]. Thus, the overall efficiency of the AGE removal system depends on renal clearance. Kidney dysfunction, as occurs in patients with nephropathy, results in failure to clear circulating AGEs and accounts, in large part, for the marked elevation of serum and tissue AGE levels observed in such patients [6, 7, 94–96]. This mechanism may also contribute to the acceleration of extrarenal vascular damage in patients with end stage renal disease (ESRD).

It is also becoming evident that intracellular protective systems exist to limit the accumulation of reactive AGE intermediates. One such system involves the degradative glyoxalase enzymes which can metabolize the reactive dicarbonyl MG to S-d-lactoylglutathione, employing reduced glutathione as a cofactor. The utility and efficiency of such systems is supported in studies in which glyoxalase-1 expression was up-regulated by gene transfection in endothelial cells, resulting in significant inhibition of AGE-mediated cell abnormalities such as increased endocytosis [97].

AGEs as mediators of diabetic vascular complications

Macroangiopathy

Diabetics are more likely to develop serious cardiovascular and cerebrovascular disease than are nondiabetics and are at increased risk for stroke and myocardial infarction caused by vascular occlusion [98–100]. Plaque formation is the most common pathophysiological hallmark of the occlusive process. The interactions between blood borne components, cytokines, growth factors, and the different vessel wall cell types which contribute to atherogenesis is extremely complex and multifactorial [101]. Atheromatous plaque formation in diabetics is indistinguishable from that occurring in nondiabetics, although the distribution of plaques may be different and diabetic lesions characteristically show a higher tendency for focal medial calcification [100]. The occurrence of fatty streaks secondary to the accumulation of lipids and lipoproteins in the vessel wall is an important early step in the evolution of advanced atherosclerotic lesions. It has been widely speculated that oxidative modification of LDL (ox-LDL) *in vivo* results in its reduced recognition by the normal LDL receptor [102, 103], leading to delayed clearance, increases in serum LDL levels [104, 105], and eventually enhanced uptake of the ox-LDL by scavenger receptors on macrophages and vascular smooth muscle cells. As mentioned earlier, advanced glycation of LDL is a physiologically relevant modification which can occur concomitantly with oxidation of LDL *in vitro* [14]. Indeed, recently, AGEs have been accepted as having an important role in the formation and acceleration of atherosclerotic lesions even in normoglycemic patients [17], but especially in diabetics [106, 107] and more so in diabetics with renal insufficiency [108].

Advanced glycation endproducts have been detected within atherosclerotic lesions in both extra- and intra-cellular locations [17, 108–111]. We have recently reported a significant correlation between serum AGE–ApoB and AGE levels in the vessel wall of carotid arteries from nondiabetic patients with occlusive disease requiring endarterectomy [17]. A similar correlation was also demonstrated between

serum AGE-LDL and AGE-immunoreactivity in vascular tissue sections from these patients [17]. Significantly, enhanced AGE-R1 and AGE-R2 were identified in cellular components of atherosclerotic lesions, with a distribution pattern consistent with enhanced tissue AGE immunoreactivity, in the same series [16, 17].

Vascular endothelium expresses receptors for AGEs [55, 58, 61, 62] and it is likely that AGE-LDL is endocytosed by the endothelium via these receptors, leading directly to the accumulation of AGEs in the subendothelial space. In addition, it has been speculated that intracellular accumulation of AGEs may promote phenotypic conversion of smooth muscle cells and foam cell formation within the atherosclerotic plaque. This is consistent with the diffuse pattern of AGE deposition and endocytosis by endothelium, smooth muscle cells and macrophages [106–112]. It also suggests that the process begins early, persists during the entire atheroma formation, and is accelerated during diabetes.

It is now understood that AGE adducts residing in the vessel wall can interfere with endothelium derived nitric oxide synthase and the vasodilatory action of nitric oxide (NO.) [113, 114]. AGEs react directly to inactivate NO-mediated vasodilatation [114] whilst AGE-infused nondiabetic animals show diabetic-like disruption of vaso-relaxation [115]. These alterations may contribute to hypertension, renal impairment and male impotence seen in diabetic patients. Endothelial dysfunction leading to enhanced procoagulant activity may result from exposure of endothelial cells to AGEs *in vitro* [116]. Furthermore, vascular endothelial cells may react to AGEs by promotion of cell adhesion and transendothelial migration [60, 117]. Thus, by causing significant dysfunctional changes in the macrovascular endothelium, AGEs can potentiate vessel wall atherogenesis, hypertension, or prothrombotic events independent of, but enhanced by, diabetes.

Microangiopathy

Diabetic microangiopathy is a broad term that describes dysfunctional changes in microvascular beds in which endothelium and associated mural cells are progressively damaged, resulting in capillary occlusion, ischaemia, and organ failure. Whilst these abnormalities are most obviously manifested

in the kidneys and retina of diabetics, microangiopathy can occur in a wide range of tissues. In fact, damage to the microvasculature in peripheral nerves is now becoming recognized as a major pathogenic factor in diabetic neuropathy [118, 119]. Recently, AGEs have been increasingly implicated in the pathogenesis of diabetic microangiopathy [120, 121]. However, their role in diabetic nephropathy and retinopathy is still under intense investigation.

Diabetic nephropathy is characterized by increased glomerular BM thickening [79] and mesangial ECM deposition, followed by mesangial hypertrophy and diffuse and nodular glomerulosclerosis [122]. Loss of glomerular function is accompanied by a reduction in filtration capacity, culminating in complete renal failure [123]. Structural changes in the glomerulus during diabetes are accompanied by several biochemical abnormalities including the accumulation of AGEs. Immunohistochemical studies of kidney from normal and diabetic rats have suggested that glomerular BM, mesangium, podocytes, and renal tubular cells accumulate high levels of AGEs. Ultrastructural studies have demonstrated BSA-AGE gold conjugate binding to glomerular structures of rats [124] and have indicated that AGE peptides may be reabsorbed by the renal proximal tubular cells [125]. AGE deposition can lead to glomerulosclerosis and widespread dysfunction independent of diabetes [126–128]. This had already been suggested by studies in which normal, nondiabetic animals were administered AGE species-specific albumin. Chronic infusion of nondiabetic animals with AGE-albumin resulted in glomerular hypertrophy, BM thickening, mesangial ECM expansion, and albuminuria, all consistent with a glomerular pathology resembling diabetic nephropathy [51, 115].

The prevalence of diabetic retinopathy in type 1 diabetic patients with T1D for more than 10 years is around 80% and diabetic retinopathy remains amongst the leading causes of blindness in the USA [129, 130]. Diabetic retinopathy is principally a disease of the intraretinal blood vessels, which become dysfunctional in response to hyperglycaemia, with progressive loss of retinal pericytes and eventually of endothelial cells, leading to capillary closure and widespread retinal ischaemia. As in other vascular beds, AGEs have been localized in the retinal vessels of diabetics [75, 131]. The precise role that

these adducts play in the pathogenesis of diabetic retinopathy remains ill-defined, although experimental studies have demonstrated that AGEs may be responsible for some retinal pathology [131] and that AGE-inhibitors, such as anti-glycated albumin [132] or aminoguanidine, can prevent the development of diabetes-associated retinal vascular lesions in rats [133] and in dogs [134]. Interestingly, aminoguanidine does not prevent the initial phase of experimental diabetic retinopathy in rats [135], although a secondary intervention study with this drug has been shown to retard disease progression [136].

Anti-AGE strategies

The determination of the mechanisms of AGE toxicity has been a key strategy in the attempt to prevent diabetic complications. To date there have been several approaches which seek to prevent AGE-formation, reduce AGE effects on cells and break pre-existing AGE crosslinks.

Amadori product formation is the basis of advanced glycation biochemistry because progression to protein crosslinks requires slow rearrangement of the Amadori to create reactive intermediates that can react with amino groups prior to the formation of irreversible AGEs. An important pharmacological strategy for the inhibition of this process has utilized the small nucleophilic hydrazine compound aminoguanidine, a potent inhibitor of AGE-mediated crosslinking [137]. The terminal amino group of aminoguanidine, by virtue of its low pKa, reacts specifically with glucose-derived reactive intermediates to prevent crosslinking. Aminoguanidine has been shown to prevent diabetes related vascular complications in experimental animals by numerous workers [14, 133, 134, 138–145]. From these extensive studies, it is apparent that aminoguanidine could be used to prevent AGE mediated tissue damage caused by diabetes and ageing. In humans, a phase I clinical trial of aminoguanidine measured advanced glycation modified haemoglobin (AGE-Hb) in treated and untreated diabetic subjects and found that AGE-Hb, as well as LDL, was significantly reduced in the treated group. HbA1c values were not affected by aminoguanidine treatment, pointing to the specificity of aminoguanidine for inhibition of post-Amadori, advanced glycation reactions. Aminoguanidine and related AGE inhibitors may eventually find widespread use in diabetics

or in individuals at risk for age related vascular sequelae. Other AGE-inhibiting drugs have been under development including the thiazolidine derivative OPB-9195 which has been shown to prevent the progression of diabetic glomerulosclerosis in rats [146].

Prevention of the interactions between AGEs and their receptors or other body proteins is a valid therapeutic approach. The use of neutralizing antibodies against glycated albumin has been shown to prevent BM thickening in diabetic db/db mice without altering the glycaemic status of the animals [132]. Likewise, the AGE-binding properties of lysozyme [70] have been used to reduce AGE levels in the dialysate from diabetic patients with kidney disease [147]. The protocol involves the capture of *in vivo*-derived AGEs with lysozyme linked to a Sepharose matrix allowing the selective depletion of AGEs from sera or dialysate [147]. This approach offers the potential for more efficient reduction of toxic AGEs in body fluids of patients with renal failure by dialysis as well as by *in vivo* lysozyme administration [72, 73].

Recently, a promising therapeutic strategy has been to attack the irreversible intermolecular AGE crosslinks formed in biological systems. This is indeed an exciting approach because it aims to 'break' preaccumulated AGEs and subsequently clear them via the kidney [148, 149]. Such an AGE crosslink 'breaker' prototype, N-phenyl-thiazolium bromide (PTB) has been described to attack covalent carbon-carbon bonds of dicarbonyl-derived crosslinks *in vitro* [148]. More recently, such an AGE-breaker named ALT-711 was found to be capable of reversing AGE-mediated vascular stiffness and distensibility in diabetic rats [150]. If equally effective in the humans, such agents may prove highly beneficial in the reversal of late cardiovascular complications that afflict older adult populations, regardless of diabetes.

References

- 1 Pirart J. Diabetes mellitus and its degenerative complications: prospective study of 4400 patients observed between 1947 and 1973. *Diabetes Care* 1978; **188**: 252–63.
- 2 The Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long term complications in insulin dependent diabetes mellitus. *N Eng J Med* 1993; **329**: 977–86.
- 3 Ruderman N, Williamson J, Brownlee M, eds. *Hyperglycemia, Diabetes and Vascular Disease*. New York, Oxford: American Physiological Society, Oxford University Press, 1992.

- 4 King GL, Kunisaki M, Nishio Y, Inoguchi T, Shiba T, Xia P. Biochemical and molecular mechanisms in the development of diabetic vascular complications. *Diabetes* 1996; **45** (Suppl. 3): S105–8.
- 5 Engerman RL, Kern TS. Progression of incipient diabetic retinopathy during good glycemic control. *Diabetes* 1987; **36**: 808–12.
- 6 Makita Z, Radoff S, Rayfield EJ, Yang Z, Skolnik E, Delaney V *et al.* Advanced glycosylation end products in patients with diabetic nephropathy. *N Engl J Med* 1991; **325**: 836–42.
- 7 Makita Z, Bucala R, Rayfield EJ, Friedman EA, Kaufman AM, Korbet SM *et al.* Reactive glycosylation endproducts in diabetic uraemia and treatment of renal failure. *Lancet* 1994; **343**: 1519–22.
- 8 Giardino I, Edelstein D, Brownlee M. Nonenzymatic glycosylation in vitro and in bovine endothelial cells alters basic fibroblast growth factor activity. A model for intracellular glycosylation in diabetes. *J Clin Invest* 1994; **94**: 110–117.
- 9 Brownlee M, Vlassara H, Cerami A. Inhibition of heparin-catalyzed human antithrombin III activity by nonenzymatic glycosylation. Possible role in fibrin deposition in diabetes. *Diabetes* 1984; **33**: 532–35.
- 10 Watkins NG, Thorpe SR, Baynes JW. Glycation of amino groups in protein. Studies on the specificity of modification of RNase by glucose. *J Biol Chem* 1985; **260**: 10629–36.
- 11 Tsilibary EC, Charonis AS, Reger LA, Wohlueter RM, Furcht LT. The effect of nonenzymatic glucosylation on the binding of the main noncollagenous NC1 domain to type IV collagen. *J Biol Chem* 1988; **263**: 4302–8.
- 12 Vitek MP, Bhattacharya K, Glendening JM, Stopa E, Vlassara H, Bucala R *et al.* Advanced glycation end products contribute to amyloidosis in Alzheimer disease. *Proc Natl Acad Sci USA* 1994; **91**: 4766–70.
- 13 Bucala R, Mitchell R, Arnold K, Innerarity T, Vlassara H, Cerami A. Identification of the major site of apolipoprotein B modification by advanced glycosylation end products blocking uptake by the low density lipoprotein receptor. *J Biol Chem* 1995; **270**: 10828–32.
- 14 Bucala R, Makita Z, Koschinsky T, Cerami A, Vlassara H. Lipid advanced glycosylation: pathway for lipid oxidation in vivo. *Proc Natl Acad Sci USA* 1993; **90**: 6434–8.
- 15 Bucala R, Makita Z, Vega G, Grundy S, Koschinsky T, Cerami A, Vlassara H. Modification of low density lipoprotein by advanced glycation end products contributes to the dyslipidemia of diabetes and renal insufficiency. *Proc Natl Acad Sci USA* 1994; **91**: 9441–5.
- 16 Stitt AW, Vlassara H, Bucala R. Atherogenesis and advanced glycation: promotion, progression, and prevention. *Annals of the New York Academy of Sciences* 1997; Vol 811. In: Chiorazzi N, Fujio-Numano RG, Ross R eds. 'Atherosclerosis IV'. New York: New York Academy of Sciences, 1997, 115–29.
- 17 Stitt AW, He C, Friedman S, Scher L, Rossi P, Ong L *et al.* Elevated AGE-modified ApoB in the sera of euglycemic, normolipidemic patients with atherosclerosis. *Molecular Medicine* 1997; **3**: 617–27.
- 18 Sell DR, Monnier VM. End-stage renal disease and diabetes catalyze the formation of a pentose-derived crosslink from aging human collagen. *J Clin Invest* 1990; **85**: 380–4.
- 19 Monnier VM, Cerami A. Nonenzymatic browning in vivo: possible process for aging of long-lived proteins. *Science* 1981; **211**: 491–3.
- 20 Kohn RR, Cerami A, Monnier VM. Collagen aging in vitro by nonenzymatic glycosylation and browning. *Diabetes* 1984; **33**: 57–9.
- 20 Harding JJ, Crabbe MJC. The lens: development proteins, metabolism and cataract. In: Davson H ed. *The Eye*, Vol 1b. New York, NY: Academic, 1984; 207–492.
- 21 Stevens VJ, Rouzer CA, Monnier VM, Cerami A. Diabetic cataract formation: potential role of glycosylation of lens crystallins. *Proc Natl Acad Sci USA* 1978; **75**: 2918–22.
- 23 Monnier VM, Stevens VJ, Cerami A. The browning reaction of proteins with glucose. *Arch Biochem* 1979; **24**: 157–78.
- 24 Matsumoto K, Ikeda K, Horiuchi S, Zhao H, Abraham EC. Immunochemical evidence for increased formation of advanced glycation end products and inhibition by amino-guanidine in diabetic rat lenses. *Biochem Biophys Res Commun* 1997; **241**: 352–4.
- 25 Stitt AW, Moore J, Sharkey JA, Murphy G, Simpson DA, Bucala R *et al.* Advanced glycation endproducts in vitreous: structural and functional implications for diabetic vitreopathy. *Invest Ophthalmol Vis Sci* 1998; **39**: 2517–23.
- 26 Charonis AS, Tsilibary EC. Structural and functional changes of laminin and type IV collagen after nonenzymatic glycation. *Diabetes* 1992; **41** (Suppl. 2): 49–51.
- 27 Knecht R, Leber R, Hasslacher C. Degradation of glomerular basement membrane in diabetes. I. Susceptibility of diabetic and nondiabetic basement membrane to proteolytic degradation of isolated glomeruli. *Res Exp Med (Berl)* 1987; **187**: 323–28.
- 28 Paul RG, Bailey AJ. Glycation of collagen: the basis of its central role in the late complications of aging and diabetes. *Int J Biochem Cell Biol* 1996; **28**: 1297–310.
- 29 Haitoglou CS, Tsilibary EC, Brownlee M, Charonis AS. Altered cellular interactions between endothelial cells and nonenzymatically glucosylated laminin/type IV collagen. *J Biol Chem* 1992; **267**: 12404–7.
- 30 Hammes HP, Weiss A, Hess S, Araki N, Horiuchi S, Brownlee M, Preissner KT. Modification of vitronectin by advanced glycation alters functional properties in vitro and in the diabetic retina. *Lab Invest* 1996; **75**: 325–38.
- 31 Tarsio JF, Reger LA, Furcht LT. Decreased interaction of fibronectin, type IV collagen, and heparin due to nonenzymatic glycation. Implications for diabetes mellitus. *Biochemistry* 1987; **26**: 1014–20.
- 32 Tarsio JF, Reger LA, Furcht LT. Molecular mechanisms in basement membrane complications of diabetes. Alterations in heparin, laminin, and type IV collagen association. *Diabetes* 1988; **37**: 532–39.
- 33 Bucala R, Model P, Cerami A. Modification of DNA by reducing sugars: a possible mechanism for nucleic acid aging and age-related dysfunction in gene expression. *Proc Natl Acad Sci USA* 1984; **81**: 105–9.
- 34 Lee AT, Cerami A. Elevated glucose 6-phosphate levels are associated with plasmid mutations in vivo. *Proc Natl Acad Sci USA* 1987; **84**: 8311–14.
- 35 Bucala R, Model P, Russel M, Cerami A. Modification of DNA by glucose 6-phosphate induces DNA rearrangements in an *Escherichia coli* plasmid. *Proc Natl Acad Sci USA* 1985; **82**: 8439–42.
- 36 Jacobson EL, Cervantes-Laurean D, Jacobson MK. ADP-ribose in glycation and glycoxidation reactions. *Adv Exp Med Biol* 1997; **419**: 371–9.

- 37 Cervantes-Laurean D, Jacobson EL, Jacobson MK. Glycation and glycooxidation of histones by ADP-ribose. *J Biol Chem* 1996; **271**: 10461–9.
- 38 Gugliucci A, Bendayan M. Histones from diabetic rats contain increased levels of advanced glycation end products. *Biochem Biophys Res Commun* 1995; **212**: 56–62.
- 39 Mills JL. Malformations in infants of diabetic mothers. *Teratology* 1982; **25**: 385–94.
- 40 Mills JL, Baker L, Goldman AS. Malformations in infants of diabetic mothers occur before the seventh gestational week. Implications for treatment. *Diabetes* 1979; **28**: 292–293.
- 41 Mills JL, Knopp RH, Simpson JL, Jovanovic-Peterson L, Metzger BE, Holmes LB *et al.* Lack of relation of increased malformation rates in infants of diabetic mothers to glycaemic control during organogenesis. *N Engl J Med* 1988; **318**: 671–6.
- 42 Sgarbieri VC, Amaya J, Tanaka M, Chichester CO. Nutritional consequences of the Maillard reaction. Amino acid availability from fructose-leucine and fructose-tryptophan in the rat. *J Nutr* 1973; **103**: 657–63.
- 43 Koschinsky T, He CJ, Mitsuhashi T, Bucala R, Liu C, Buenting C *et al.* Orally absorbed reactive glycation products (glycotoxins): an environmental risk factor in diabetic nephropathy. *Proc Natl Acad Sci USA* 1997; **94**: 6474–9.
- 44 Zheng F, He C, Li J, Vlassara H. Restriction of AGE content of food without lowering protein intake prevents diabetic nephropathy in mice. *Diabetes* 2000; **49**(Suppl.): A161 (#662-P).
- 45 Lin R-Y, Dore A, Reis ED, Lu W, Fisher E, Vlassara H *et al.* Lowering of diet-derived advanced glycation endproducts reduces postangioplasty restenosis in apo-E deficient mice: a new solution? Abstract. *AHA Arterioscl Thromb & Vasc Biol* 2001; **21**: 645.
- 46 Lin Y-R, Choudhury R, Lu M, Dore A, Fisher E, Vlassara *et al.* Suppression of atherosclerotic lesions by dietary restriction of advanced glycation endproducts in diabetic Apo-E deficient C57BL/6J mice. *Diabetes* 2001; **50** (Suppl. 2): A8.
- 47 Vlassara H, Cai W, Crandall J, Goldberg T, Oberstein R, Dardaine V, Peppas M, Rayfield E. Inflammatory markers are induced by dietary glycotoxins: a pathway for accelerated atherosclerosis in diabetes (in press).
- 48 Cerami C, Founds H, Nicholl I, Mitsuhashi T, Giordano D, Vanpatten S *et al.* Tobacco smoke is a source of toxic reactive glycation products. *Proc Natl Acad Sci USA* 1997; **94**: 13915–20.
- 49 Founds HW, Giordano D, Mitsuhashi T, Stitt AW, Finch G, Cerami A *et al.* Tobacco smoke is a source of advanced glycation endproducts (AGEs): possible role in the accelerated vascular disease of smokers. *J Investigative Medicine* 1996; **44**: A200. (Abstracts)
- 50 Nicholl ID, Stitt AW, Moore JE, Ritchie AJ, Archer DB, Bucala R. Cigarette smoke-derived advanced glycation endproducts: role in cataractogenesis and atherosclerosis. *Molecular Medicine* (in press).
- 51 Vlassara H, Bucala R, Striker L. Pathogenic effects of advanced glycosylation endproducts: biochemical, biologic, and clinical implications for diabetes and aging. *Lab Invest* 1994; **70**: 138–51.
- 52 Vlassara H, Brownlee M, Cerami A. Novel macrophage receptor for glucose-modified proteins is distinct from previously described scavenger receptors. *J Exp Med* 1986; **164**: 1301–9.
- 53 Radoff S, Cerami A, Vlassara H. Isolation of surface binding protein specific for advanced glycosylation end products from mouse macrophage-derived cell line RAW 264.7. *Diabetes* 1990; **39**: 1510–18.
- 54 Yang Z, Makita Z, Horii Y, Brunelle S, Cerami A, Sehajpal P, Suthanthiran M *et al.* Two novel rat liver membrane proteins that bind advanced glycosylation endproducts: relationship to macrophage scavenger receptor for glucose modified proteins. *J Exp Med* 1991; **174**: 515–24.
- 55 Li YM, Mitsuhashi T, Wojciechowicz D, Shimizu N, Li J, Stitt A *et al.* Molecular identity and cellular distribution of advanced glycation endproduct receptors. Relationship of p60 to OST-48 and 80K-H membrane proteins. *Proceedings of the National Academy of Science, USA* 1996; **93**: 11047–52.
- 56 Goh KC, Lim YP, Ong SH, Siak CB, Cao X, Tan YH, Guy GR. Identification of p90, a prominent tyrosine-phosphorylated protein in fibroblast growth factor stimulated cells, as 80KH. *J Biol Chem* 1996; **271**: 5832–8.
- 57 Vlassara H, Li YM, Imani F, Wojciechowicz D, Yang Z, Liu FT, Cerami A. Identification of galectin-3 as a high-affinity binding protein for advanced glycation end products (AGE): a new member of the AGE-receptor complex. *Molecular Medicine* 1995; **1**: 634–46.
- 58 Stitt AW, He C, Burke G, Li YM, Vlassara H. The advanced glycation endproduct (AGE) receptor complex: characterization in micro and macrovascular endothelium. *Diabetes* 1997; **46**: 53A.
- 59 Vlassara H, Brownlee M, Manogue KR, Dinarello CA, Pasagian A. Cachectin/TNF and IL-1 induced by glucose-modified proteins: role in normal tissue remodeling. *Science* 1988; **240**: 1546–8.
- 60 Kirstein M, Aston C, Hintz R, Vlassara H. Receptor-specific induction of insulin-like growth factor I in human monocytes by advanced glycosylation end product-modified proteins. *J Clin Invest* 1992; **90**: 439–46.
- 61 Schmidt AM, Yan SD, Wautier JL, Stern D. Activation of receptor for advanced glycation end products: a mechanism for chronic vascular dysfunction in diabetic vasculopathy and atherosclerosis. *Circ Res* 1999; **84**: 489–97.
- 62 Schmidt AM, Hasu M, Popov D, Zhang JH, Chen J, Yan SD *et al.* Receptor for advanced glycation end products (AGEs) has a central role in vessel wall interactions and gene activation in response to circulating age proteins. *Proc. Natl. Acad. Sci. USA* 1994; **91**: 8807–11.
- 63 Takata K, Horiuchi S, Araki N, Shiga M, Saitoh M, Morino Y. Endocytic uptake of nonenzymatically glycosylated proteins is mediated by a scavenger receptor for aldehyde-modified proteins. *J Biol. Chem* 1988; **263**: 14819–25.
- 64 Araki N, Higashi T, Mori T, Shibayama R, Kawabe Y, Kodama T *et al.* Macrophage scavenger receptor mediates the endocytic uptake and degradation of advanced glycation end products of the Maillard reaction. *Eur J Biochem* 1995; **230**: 408–15.
- 65 Ohgami N, Nagai R, Ikemoto M, Arai H, Kuniyasu A, Horiuchi S *et al.* CD36, a member of class B scavenger receptor family, as a receptor for advanced glycation end products. *J Biol Chem* 2000; **276**: 3195–202.
- 66 Mackic J, Stins M, McComb J, Calero M, Ghiso J, Kin K *et al.* Human blood-brain barrier receptors for Ab 1–40. *J Clin Invest* 102: 734–43.

- 67 Sano H, Higashi T, Matsumoto K, Melkko J, Jinnouchi Y, Ikeda K *et al.* Insulin enhances macrophage scavenger receptor-mediated endocytic uptake of advanced glycation end products. *J Biol Chem* 1998; **273**: 8630–37.
- 68 Horiuchi, S, Higashi T, Ikeda K, Saishoji T, Jinnouchi, Sano H *et al.* Advanced glycation endproducts and their recognition by macrophage and macrophage-derived cells. *Diabetes* 1996; **45** (Suppl 3): S73–6.
- 69 Sava G. Pharmacological aspects and therapeutic applications of lysozymes. In: Jolies P ed. *Lysozymes: Model Enzymes in Biochemistry and Biology*. Basel, Switzerland: Birkhauser Verlag, 1996; 433–49.
- 70 Li YM, Tan AX, Vlassara H. Antibacterial activity of lysozyme and lactoferrin is inhibited by binding of advanced glycation-modified proteins to a conserved motif. *Nat Med* 1995; **1**: 1057–61.
- 71 Mitsuhashi T, Li YM, Fishbane S, Vlassara H. Depletion of reactive advanced glycation endproducts from diabetic uremic sera by a lysozyme-linked matrix. *J Clin Invest* 1997; **100**: 847–54.
- 72 Zheng F, Cai W, Mitsuhashi T, Vlassara H. Lysozyme enhances renal excretion of advanced glycation endproducts *in vivo* and suppresses adverse AGE-mediated cellular effects *in vitro*: a potential AGE sequestration therapy for diabetic nephropathy? *Mol Med* (in press).
- 73 Zheng F, He C, Li J, Vlassara H. *In vivo* treatment of diabetic mice (NOD and db/db) with lysozyme (LZ) enhances renal clearance of glycoxidation products and prevents albuminuria. *Diabetes* 2000; **49**: A161.
- 74 Vlassara H, Li J, Bohmart A, Zheng F. Lysozyme (LZ) is secreted by mesangial cells and inhibits glycoxidation-dependent glomerular gene dysregulation *in vitro* and *in vivo*: a novel defense mechanism against diabetic nephropathy. *Diabetes* 2000; **49**: A161.
- 75 Stitt AW, Li YM, Gardiner TA, Bucala R, Archer DB, Vlassara H. Advanced glycation endproducts (AGEs) co-localize with AGE receptors in the retinal vasculature of diabetic and of AGE-infused rats. *Am J Pathol* 1997; **150**: 523–31.
- 76 Vlassara H, Brownlee M, Cerami A. Specific macrophage receptor activity for advanced glycosylation end products inversely correlates with insulin levels *in vivo*. *Diabetes* 1988; **37**: 456–61.
- 77 Vlassara H, Moldawer L, Chan C. Macrophage/monocyte receptor for nonenzymatically glycosylated protein is upregulated by cachectin/tumor necrosis factor. *J Clin Invest* 1989; **84**: 1813–20.
- 78 Vlassara H. Advanced non-enzymatic tissue glycosylation: mechanisms implicated in the complications associated with aging. In: Clegg M, O'Brien S eds. *Molecular Biology of Aging*. New York: Alan R. Liss, Inc., 1989; 171–85.
- 79 Radoff S, Vlassara H, Cerami A. Characterization of a solubilized cell surface binding protein on macrophages specific for proteins modified nonenzymatically by advanced glycosylated end products. *Arch Biochem Biophys* 1998; **263**: 418–23.
- 80 Stitt, AW, Burke G, Chen F, McMullen CBT, Vlassara H. Advanced glycation end product receptor interactions on microvascular cells occur within caveolin-rich membrane domains. *FASEB J* 2000; **14**: 2390–2, 10/1096/fj.00–0289fje. (express article)
- 81 Pugliese G, Pricci F, Iacobini C, Leto G, Amadio L, Barsotti P *et al.* Accelerated diabetic glomerulopathy in Galectin-3/AGE-receptor-3 knockout mice. *FASEB* 2001; **15**: 2471–9.
- 82 He C, Zheng F, Stitt A, Striker L, Masakazu H, Vlassara H. Differential expression of renal AGE-receptor genes in NOD mice: possible role in NOD diabetic renal disease. *Kidney International* 2000; **58**: 1931–40.
- 83 He C, Stitt A, Striker L, Hatori M, Vlassara H. Low expression of AGE-receptor-1 in NOD mouse mesangial cells: possible link to diabetic nephropathy. *J Am Soc Nephrol* 1996; **7**: 1871A
- 84 He C, Koschinsky T, Buening C, Vlassara H. Presence of diabetic complications in type 1 diabetic patients correlates with low expression of mononuclear AGE-receptor-1 and elevated serum AGE. *Molec Med* 2001; **7**: 159–168.
- 85 He C, Koschinsky T, Sabol J, Buening C, Liu C, Vlassara H. Mononuclear (MN) cell AGE receptor-1 (AGE-R1) mRNA expression and its relationship to diabetic complications. *Diabetes* 1997; **46**:8A.
- 86 Kadrofske M, Openo KP, Wang JL. The human LGALS3 (Galectin-3) gene: determination of the gene structure and functional characterization of the promoter. *Archives Biochem and Biophys* 1998; **349**: 7–20.
- 87 Landers HM, Tauras JM, Ogiste JS. Isolation of cDNAs encoding a substate for protein kinase C: nucleotide sequence and chromosomal mapping of the gene for a human 80K-H protein. *J Biol Chem* 1989; **272**: 17810–14.
- 88 Yamagata T, T. Tsuru T, Momoi M, Suwa K, Nozaki Y, Mukasa *et al.* Genome organization of human 48-kDa oligosaccharyltransferase (DDOST). *Genomics* 1997; **45**: 535–40.
- 89 Poirier O, Nicaud V, Vionnet N, Raoux S, Tarnow L, Vlassara H *et al.* Polymorphism screening of four genes encoding advanced glycation end product putative receptors: association study with nephropathy in Type 1 diabetes patients. *Diabetes* 2001; **50**: 1214–18.
- 90 Hudson BI, Strickland MH, Grant PJ. Identification of polymorphisms in the receptor for advanced glycation end products (RAGE) gene. *Diabetes* 1998; **47**: 1155–7.
- 91 Gardiner TA, Stitt AW, Archer DB. Endocytosis by retinal vascular endothelial cells increases in early diabetes: a quantitative EM study in STZ-diabetic rats. *Laboratory Investigation* 1995; **72**: 439–44.
- 92 Stitt AW, Chakravarthy U, Archer DB, Gardiner TA. Increased endocytosis of retinal vascular endothelial cells grown in hyperglycemia is modulated by inhibitors of non-enzymatic glycosylation. *Diabetologia* 1995; **38**: 1271–1275.
- 93 Vlassara H. Recent progress in advanced glycation end products and diabetic complications. *Diabetes* 1997; **46**: (Suppl 2): S19–25.
- 94 Miyata T, Oda O, Inagi R, Iida Y, Araki N, Yamada N *et al.* beta 2-Microglobulin modified with advanced glycation end products is a major component of hemodialysis-associated amyloidosis. *J Clin Invest* 1993; **92**: 1243–52.
- 95 Miyata T, Inagi R, Iida Y, Sato M, Yamada N, Oda O *et al.* Involvement of beta 2-microglobulin modified with advanced glycation end products in the pathogenesis of hemodialysis-associated amyloidosis. Induction of human monocyte chemotaxis and macrophage secretion of tumor necrosis factor-alpha and interleukin-1. *J Clin Invest* 1994; **93**: 521–8.
- 96 Dolhofer-Bliesener R, Lechner B, Gerbitz KD. Possible significance of advanced glycation end products in serum in end-stage renal disease and in late complications of diabetes. *Eur J Clin Chem Clin Biochem* 1996; **34**: 355–61.

- 97 Shinohara M, Thornalley PJ, Giardino I, Beisswenger P, Thorpe SR, Onorato J, Brownlee M. Overexpression of glyoxalase-1 in bovine endothelial cells inhibits intracellular advanced glycation endproduct formation and prevents hyperglycemia-induced increases in macromolecular endocytosis. *J Clin Invest* 1998; **101**: 1142–7.
- 98 Lithner F, Asplund K, Eriksson S, Hagg E, Strand T, Wester PO. Clinical characteristics in diabetic stroke patients. *Diabete Metab* 1988; **14**: 15–19.
- 99 Pyorala K. Diabetes and coronary artery disease: what a coincidence? *J Cardiovasc Pharmacol* 1990; **16**: S8–14.
- 100 Ruderman NB, Haudenschild C. Diabetes as an atherogenic factor. *Progress in cardiovascular Diseases* 1984; **26**: 273–412.
- 101 Ross R. The pathogenesis of atherosclerosis: a perspective for the 1990s. *Nature* 1993; **362**: 801–9.
- 102 Mahley RW, Innerarity TL, Weisgraber KN, Oh SY. Altered metabolism (in vivo and in vitro) of plasma lipoproteins after selective chemical modification of lysine residues of the apoprotein B. *J Clin Invest* 1979; **64**: 743–50.
- 103 Mahley RW, Innerarity TL, Pitas RE, Weisgraber KH, Brown JH, Gross E. Inhibition of lipoprotein binding to surface receptors of fibroblasts following selective modification of arginyl residues in apoprotein B. *J Biol Chem* 1987; **252**: 7279–87.
- 104 Goldstein JL, Ho YK, Basu SK, Brown MS. Binding site on macrophages that mediates uptake and degradation of acetylated low density lipoproteins producing massive cholesterol deposition. *Proc Natl Acad Sci USA* 1979; **76**: 333–7.
- 105 Fogelman AM, Haberland ME, Seager J, Hokom M, Edwards PA. Factors regulating the activities of the low density lipoprotein receptor and the scavenger receptor on human monocytes macrophages. *J Lipid Res* 1980; **22**: 1131–41.
- 106 Palinski W, Koschinsky T, Butler SW, Miller E, Vlassara H, Cerami A, Witztum JL. Immunological evidence for the presence of advanced glycosylation endproducts in atherosclerotic lesions of euglycemic rabbits. *Arterioscler Thromb Vasc Biol* 1995; **15**: 571–82.
- 107 Kume S, Takeya M, Mori T, Araki N, Suzuki H, Horiuchi S *et al.* Immunohistochemical and ultrastructural detection of advanced glycation endproducts in atherosclerotic lesions of human aorta with a novel specific monoclonal antibody. *Am J Pathol* 1995; **147**: 654–67.
- 108 Nakamura Y, Horii Y, Nishino T, Shiiki H, Sakaguchi Y, Kagoshima T *et al.* Immunohistochemical localization of advanced glycosylation endproducts (AGEs) in coronary atheroma and cardiac tissue in diabetes. *Am J Pathol* 1993; **143**: 1649–56.
- 109 Sima A, Popov D, Starodub O, Stancu C, Cristea C, Stern D, Simionescu M. Pathobiology of the heart in experimental diabetes: immunolocalization of lipoproteins, immunoglobulin G, and advanced glycation endproducts proteins in diabetic and/or hyperlipidemic hamster. *Lab Invest* 1997; **77**: 3–18.
- 110 Niwa T, Katsuzaki T, Miyazaki S, Miyazaki T, Ishizaki Y, Hayasi F *et al.* Immunohistochemical detection of imidazolone, a novel advanced glycation end product, in kidneys and aortas of diabetic patients. *J Clin Invest* 1997; **99**: 1272–80.
- 111 Horiuchi S, Sano H, Higashi T, Ikeda K, Jinnouchi Y, Nagai R, Takahashi K. Extra- and intracellular localization of advanced glycation end-products in human atherosclerotic lesions. *Nephrol Dial Transplant* 1996; **11** (Suppl 5): 81–6.
- 112 Dobrian A, Lazar V, Tirziu D, Simionescu M. Increased macrophage uptake of irreversibly glycated albumin modified-low density lipoproteins of normal and diabetic subjects is mediated by non-saturable mechanisms. *Biochim Biophys Acta* 1996; **1317**: 5–14.
- 113 Seftel AD, Vaziri ND, Ni Z, Razmjouei K, Fogarty J, Hampel N *et al.* Advanced glycation end products in human penis: elevation in diabetic tissue, site of deposition, and possible effect through iNOS or eNOS. *Urology* 1997; **50**: 1016–26.
- 114 Bucala R, Tracey KJ, Cerami A. Advanced glycosylation products quench nitric oxide and mediate defective endothelium dependent vasodilatation in experimental diabetes. *J Clin Invest* 1991; **87**: 432–8.
- 115 Vlassara H, Fuh H, Makita Z, Krunkkrai S, Cerami A, Bucala R. Exogenous advanced glycosylation end products induce complex vascular dysfunction in normal animals: a model for diabetic and ageing complications. *Proc Natl Acad Sci USA* 1992; **89**: 12043–7.
- 116 Stern DM, Esposito C, Gerlach H, Gerlach M, Ryan J, Handley D, Nawroth P. Endothelium and regulation of coagulation. *Diabetes Care* 1991; **14**: 160–6.
- 117 Vlassara H, Fuh H, Donnelly T, Cybulsky M. Advanced glycation endproducts promote adhesion molecule (VCAM-1, ICAM-1) expression and atheroma formation in normal rabbits. *Mol Med* 1995; **1**: 447–56.
- 118 Dyck PJ, Hansen S, Karnes J, O'Brien P, Yasuda H, Windebank A, Zimmerman B. Capillary number and percentage closed in human diabetic sural nerve. *Proc Natl Acad Sci USA* 1985; **82**: 2513–7.
- 119 Vinik AI, Holland MT, Le Beau JM, Liuzzi FJ, Stansberry KB, Colen LB. Diabetic neuropathies. *Diabetes Care* 1992; **15**: 1926–75.
- 120 Soulis Liparota T, Cooper M, Papazoglou D, Clarke B, Jerums G. Retardation by aminoguanidine of development of albuminuria, mesangial expansion, and tissue fluorescence in streptozotocin induced rat. *Diabetes* 1991; **40**: 1328–34.
- 121 La Selva M, Beltramo E, Passera P, Porta M, Molinatti GM. The role of endothelium in the pathogenesis of diabetic microangiopathy. *Acta Diabetol* 1993; **30**: 190–200.
- 122 Osterby R, Anderson MJF, Gundersen HJG, Jorgensen HE, Mogensen CE, Parving HH. Quantitative study on glomerular ultrastructure in type I diabetes with incipient nephropathy. *Diab Nephrop* 1983; **3**: 95–100.
- 123 Mogensen CE. Renal function changes in diabetes. *Diabetes* 1976; **25** (Suppl. 2): 872–9.
- 124 Gugliucci A, Bendayan M. Reaction of advanced glycation endproducts with renal tissue from normal and streptozotocin induced diabetic rats. An ultrastructural study using colloidal gold cytochemistry. *J Histochem Cytochem* 1995; **43**: 591–600.
- 125 Gugliucci A, Bendayan M. Renal fate of circulating advanced glycation end products (AGE): evidence for reabsorption and catabolism of AGE peptides by renal proximal tubular cells. *Diabetologia* 1995; **39**: 149–160.
- 126 Makino H, Shikata K, Hironaka K, Kushihiro M, Yamasaki Y, Sugimoto H *et al.* Ultrastructure of nonenzymatically glycosylated mesangial matrix in diabetic nephropathy. *Kidney Int* 1995; **48**: 517–26.
- 127 Pugliese G, Pricci F, Romeo G, Pugliese F, Mene P, Giannini S *et al.* Upregulation of mesangial growth factor and extracellular matrix synthesis by advanced glycation end

- products via a receptor-mediated mechanism. *Diabetes* 1997; **46**: 1881–7.
- 128 Yamauchi A, Takei I, Makita Z, Nakamoto S, Ohasi N, Kiguchi H *et al.* Effects of aminoguanidine on serum advanced glycation endproducts, urinary albumin excretion, mesangial expansion, and glomerular basement membrane thickening in Otsuka Long-Evans Tokushima fatty rats. *Diabetes Res Clin Pract* 1997; **34**: 127–33.
 - 129 Kahn HA, Moorhead HB. *Statistics on Blindness in the Model Reporting Area, 1969–1970*. U.S. Department of Health, Education, and Welfare Publication No. (NIH) 73–427. Washington: U.S. Government Printing Office, 1973.
 - 130 Brownlee M, Vlassara H, Cerami A. Advanced glycosylation end products in tissue and the biochemical basis of diabetic complications. *N Eng J Med* 1988; **318**: 1315–1321.
 - 131 Hammes HP, Wellensiek B, Kloting I, Sickel E, Bretzel RG, Brownlee M. The relationship of glycaemic level to advanced glycation end-product (AGE) accumulation and retinal pathology in the spontaneous diabetic hamster. *Diabetologia* 1998; **41**: 165–70.
 - 132 Clements RS Jr, Robison WG Jr, Cohen MP. Anti-glycated albumin therapy ameliorates early retinal microvascular pathology in db/db mice. *J Diabetes Comp* 1998; **12**: 28–33.
 - 133 Hammes HP, Martin S, Federlin K, Geisen K, Brownlee M. Aminoguanidine treatment inhibits the development of experimental diabetic retinopathy. *Proc Natl Acad Sci USA* 1991; **88**: 11555–8.
 - 134 Kern TS, Engerman RL. Pharmacological inhibition of diabetic retinopathy. *Diabetes* 2001; **50**: 1636–42.
 - 135 Hammes HP, Ali SS, Uhlmann M, Weiss A, Federlin K, Geisen K, Brownlee M. Aminoguanidine does not inhibit the initial phase of experimental diabetic retinopathy in rats. *Diabetologia* 1995; **38**: 269–73.
 - 136 Hammes HP, Stodter D, Weiss A, Bretzel RG, Federlin K, Brownlee M. Secondary Intervention with aminoguanidine retards the progression of diabetic retinopathy in the rat model. *Diabetologia* 1995; **38**: 656–60.
 - 137 Brownlee M, Vlassara H, Kooney A, Ulrich P, Cerami A. Aminoguanidine prevents diabetes-induced arterial wall protein cross-linking. *Science* 1986; **232**: 1629–32.
 - 138 Giardino I, Fard AK, Hatchell DL, Brownlee M. Aminoguanidine inhibits reactive oxygen species formation, lipid peroxidation, and oxidant-induced apoptosis. *Diabetes* 1998; **47**: 1114–20.
 - 139 Panagiotopoulos S, O'Brien RC, Bucala R, Cooper ME, Jerums G. Aminoguanidine has an anti-atherogenic effect in the cholesterol-fed rabbit. *Atherosclerosis* 1998; **136**: 125–31.
 - 140 Nichols K, Mandel TE. Advanced glycosylation endproducts in experimental murine diabetic nephropathy: effect of islet isografting and of aminoguanidine. *J Lab Invest* 1989; **60**: 486–91.
 - 141 Ellis EN, Good BH. Prevention of glomerular basement membrane thickening by aminoguanidine in experimental diabetes mellitus. *Metabolism* 1991; **40**: 1016–19.
 - 142 Ido Y, Chang K, Ostrow E, Allison W, Kilo C, Tilton RG. Aminoguanidine prevents regional blood flow increases in streptozotocin-diabetic rats. *Diabetes* 1990; **39**: 93A.
 - 143 Soulis Liparota T, Cooper M, Papazoglou D, Clarke B, Jerums G. Retardation by aminoguanidine of development of albuminuria, mesangial expansion, and tissue fluorescence in streptozotocin induced rat. *Diabetes* 1991; **40**: 1328–34.
 - 144 Edelstein D, Brownlee M. Mechanistic studies of advanced glycosylation endproduct inhibition by aminoguanidine. *Diabetes* 1992; **41**: 26–8.
 - 145 Cho HK, Kozu H, Peyman GA, Parry GJ, Khoobehi B. The effect of aminoguanidine on the blood retinal barrier in streptozotocin induced diabetic rats. *Ophthalmic Surg* 1991; **22**: 44–47.
 - 146 Nakamura S, Makita Z, Ishikawa S, Yasumura K, Fujii W, Yanagisawa K *et al.* Progression of nephropathy in spontaneous diabetic rats is prevented by OPB-9195, a novel inhibitor of advanced glycation. *Diabetes* 1997; **46**: 895–9.
 - 147 Mitsuhashi T, Li YM, Fishbane S, Vlassara H. Depletion of reactive advanced glycation endproducts from diabetic uremic sera using a lysozyme-linked matrix. *J Clin Invest* 1997; **100**: 847–54.
 - 148 Vasan S, Zhang X, Zhang X, Kapurniotu A, Bernhagen J, Teichberg S *et al.* An agent cleaving glucose-derived protein crosslinks in vitro and in vivo. *Nature* 1996; **382**: 275–8.
 - 149 Drickamer K. Breaking the curse of the AGEs. *Nature* 1996; **382**: 211–2.
 - 150 Wollfenbuetel BH, Boulanger CM, Crijns FR, Huilberts MS, Poitevin P, Swennen GN *et al.* Breakers of advanced glycation end products restore large artery properties in experimental diabetes. *Proc Natl Acad Sci USA* 1998; **95**: 4630–4.

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